

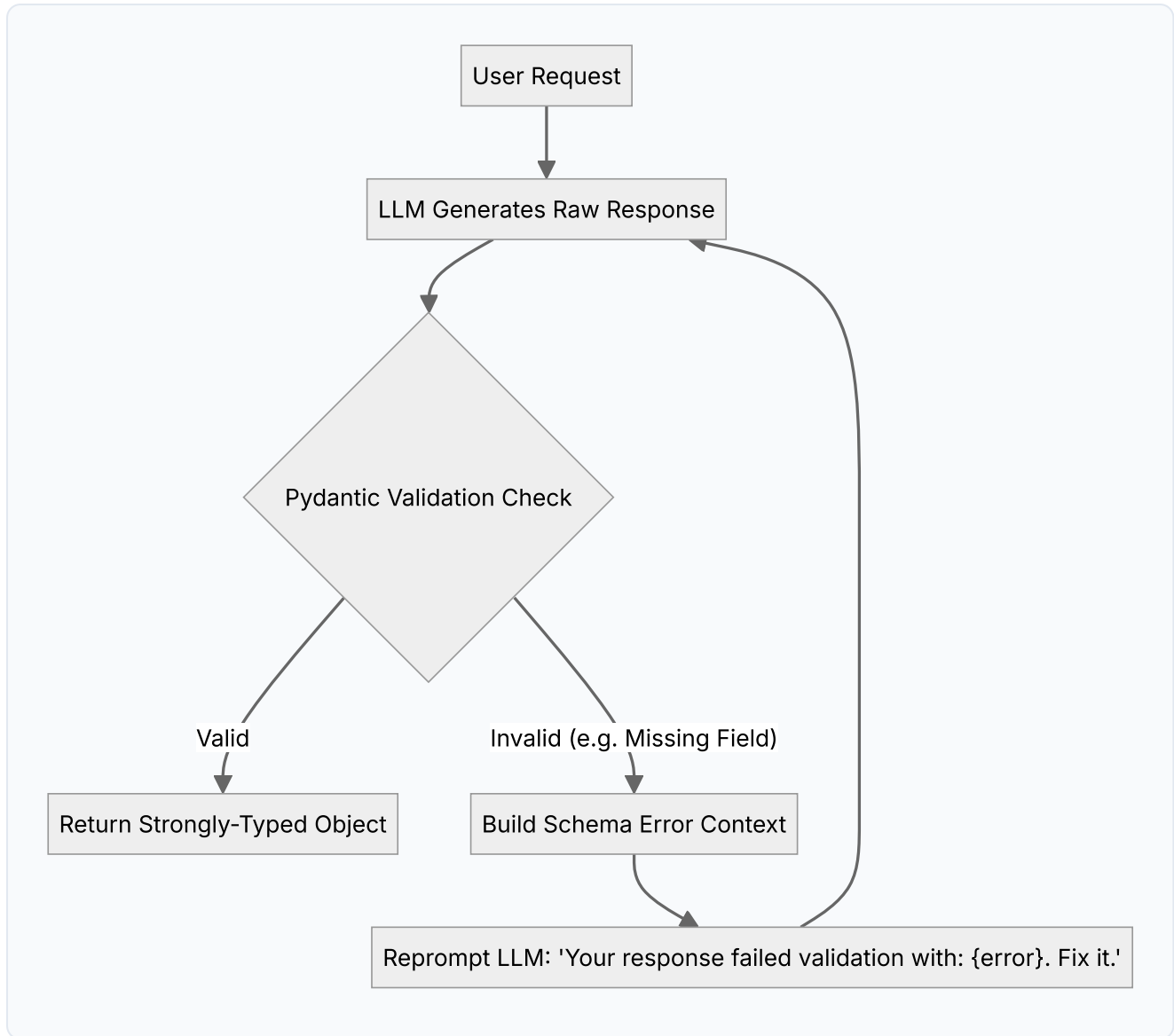
Chapter 10: Production Resilience, Defensive Prompting, & Rate Limits

The Non-Deterministic Nightmare Writing an LLM agent script in a Jupyter Notebook that succeeds once is easy. Running an agent that executes 100,000 workflows a day across enterprise customers without silently failing, hallucinating corrupt JSON, or draining your credit card via runaway recursive loops is an entirely different engineering discipline.

Production Agentic AI engineering is **defensive engineering**. This chapter covers schema healing, token bucket rate limiters with jitter, multi-provider fallback cascades, and circuit breakers.

1. Schema Enforcement & Healing with Pydantic & Instructor

LLMs frequently generate invalid JSON: trailing commas, unescaped quotes, or markdown code fences (````json`) wrapped around payloads.



Self-Healing Extraction Pipeline

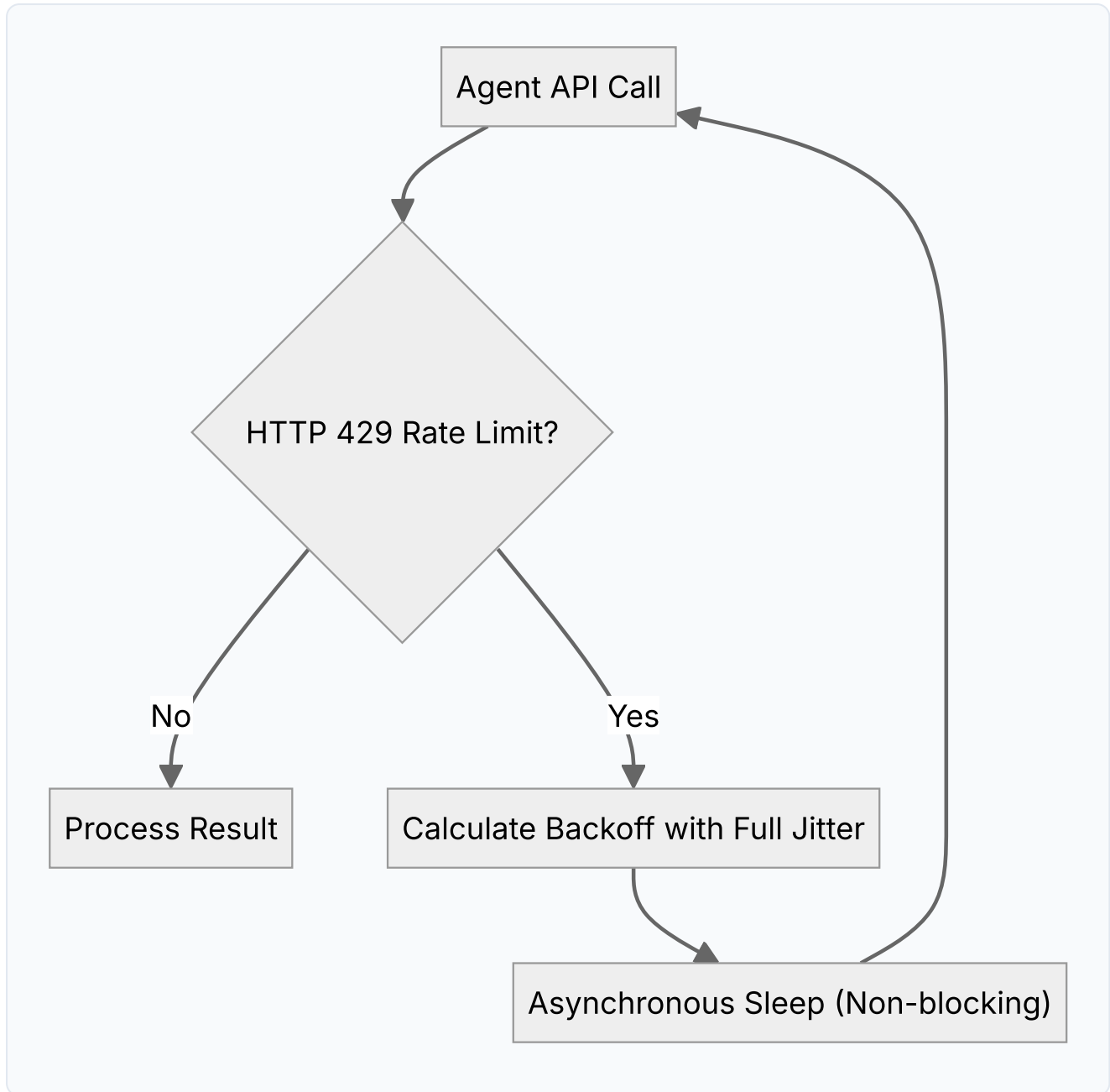
2. Rate Limits, Throttling, & Full Jitter Backoff

When you scale to 50 concurrent agents, OpenAI or Anthropic will immediately return `HTTP 429 Too Many Requests`.

Why Exponential Backoff Without Jitter Causes Avalanches

If 50 agents hit a rate limit at the exact same millisecond and all sleep for $2^1 = 2$ seconds, they will all wake up at the exact same millisecond and collide again!

$$\text{Sleep Time} = \text{random}(0, \min(M, B \times 2^{\text{attempt}}))$$

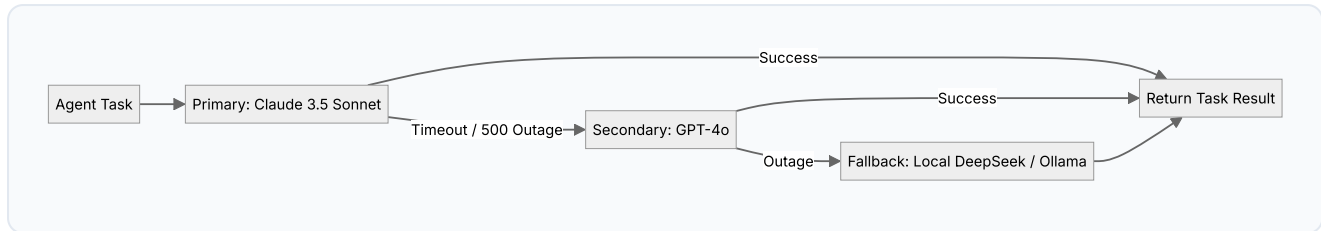


Production Rate Limiter with Full Jitter



3. Multi-Provider Fallback Cascade

Relying on a single LLM provider in production is an existential risk. When OpenAI has an API outage, your business stops. A resilient agent architecture uses a **Cascade**:



Cascade Implementation

4. Runaway Recursive Loop Circuit Breaker

An autonomous agent with tools can easily enter an infinite loop:

- Agent runs `search("weather in London")`
- Tool returns error: `KeyError: temp`
- Agent repeats `search("weather in London")` forever, consuming \$500 in tokens in 20 minutes.

The Strict Budget Token & Step Circuit Breaker

Every production agent loop must wrap its execution inside this circuit breaker!

Further Reading

- [AWS: exponential backoff and jitter](#)
- [OWASP Top 10 for LLM applications](#)
- [Martin Fowler: circuit breaker](#)

Check Yourself

Answer in your head or on paper first, then open each answer.

- ▶ 1. What does exponential backoff with jitter avoid?
- ▶ 2. Why set timeouts on every LLM and tool call?
- ▶ 3. What is defensive prompting?
- ▶ 4. What is a circuit breaker?